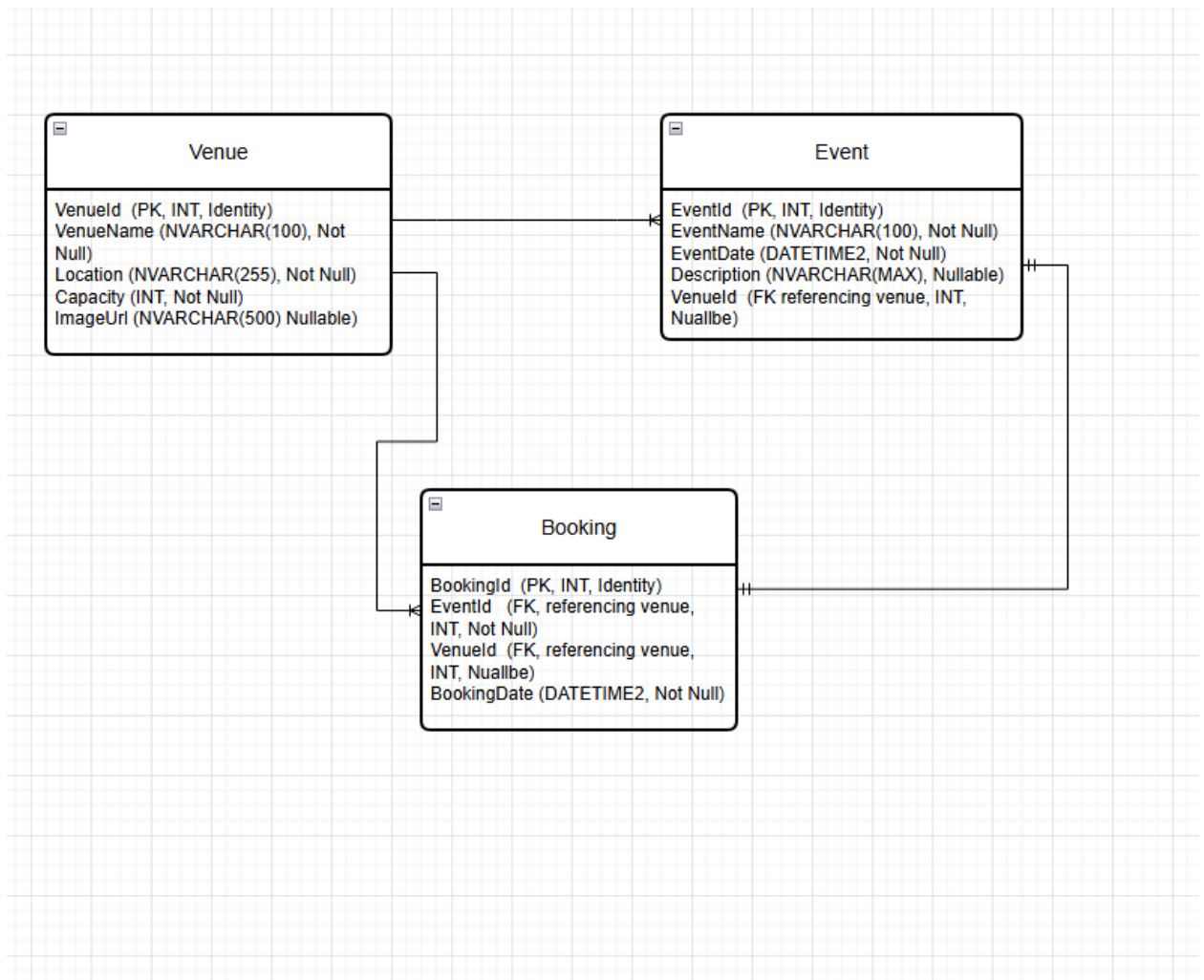


CLDV7111_POE_Part1

ST10388202

1. EventEase Database Design:



2. Deployment To Azure:

2.1 Deployed Web Application:

Venues

[Create New Venue](#)

Venue Name	Location	Capacity	Image URL	Actions
moses mabhida	durban	1222	 moses mabhida	Edit Details Delete

Edit Venue

Venue Name

Location

Capacity

Image URL

Save

Back to List

Events

Create New Event

Event Name	Event Date	Description	Venue	Actions
SummerBreeze	3/26/2026 2:43:00 PM	svcsgrfdkj	moses mabhida	Edit Details Delete

Edit Event

Event Name

SummerBreeze

Event Date

03/26/2026 02:43 PM



Description

svcsgrfdkj

Venue

moses mabhida

[Save](#)

[Back to List](#)

← → ↻ eventeaseapp-h0ejgrb9eba3akew.austriaeast-01.azurewebsites.net/Bookings/Create

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EventEaseApp Home Privacy Venues Events Bookings

Create Booking

Event

-- Select Event --

Venue

-- Select Venue --

Booking Date

mm/dd/yyyy --:-- -- 

[Create](#) [Back to List](#)

URL Link:

eventeaseapp-h0ejgrb9eba3akew.austriaeast-01.azurewebsites.net

2.2 Azure App Service Resource

portal.azure.com/#@advtechonline.onmicrosoft.com/resource/subscriptions/06e4c813-d342-49ef-9e3a-3315fd8afd6d/resourceGroups/EventEaseApp-rg/providers/Microsoft.Web/sites/EventEaseApp/appServices

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EventEaseApp

Web App

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 - Configuration
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 - Custom domains
 - Certificates
 - Networking
 - WebJobs
 - MySQL In App
 - Service Connector

You have multiple notifications. Switch to Notifications tab to get more details.

Essentials

Resource group (move)	: EventEaseApp-rg	Default domain	: eventeaseapp-h0ejgrb9eba3akew.austriaeast-01.azurewebsites.net
Status	: Running	App Service Plan	: EventEaseAppPlan
Location (move)	: Austria East	Operating System	: Windows
Subscription (move)	: Azure for Students	Health Check	: Not Configured
Subscription ID	: 06e4c813-d342-49ef-9e3a-3315fd8afd6d		
Tags (edit)	: Add tags		

Properties

Web app	
Name	EventEaseApp
Publishing model	Code
Runtime Stack	Dotnet - v8.0

Domains

Default domain	eventeaseapp-h0ejgrb9eba3akew.austriaeast-01.azurewebsites.net
Custom domain	Add custom domain

Hosting

Plan Type	App Service plan
Name	EventEaseAppPlan
Operating System	Windows

Deployment Center

Deployment logs	View logs
Last deployment	Loading deployments...
Deployment provider	None

Application Insights

Name	Enable Application Insights
------	-----------------------------

Networking

Virtual IP address	68.210.171.0
Outbound IP addresses	68.210.130.71, 68.210.171.0
Additional Outbound IP addresses	68.210.129.147, 68.210.129.167, 68.210... Show More
Virtual network integration	Not supported

Add or remove favorites by pressing **Ctrl+Shift+F**

2.3 Azure SQL Database

portal.azure.com/#@advtechonline.onmicrosoft.com/resource/subscriptions/06e4c813-d342-49ef-9e3a-3315fd8afd6d/resourceGroups/EventEaseApp-rg/providers/Microsoft.Sql/servers/eventeaseapp-server-db/databases/CLDV7111_database/overview

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CLDV7111_database (eventeaseapp-server-db/CLDV7111_database)

SQL database

Show me performance metrics for this SQL database. Suggest missing index Find query store timeouts

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- Mirror database in Fabric (preview)
- Resource visualizer
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 - Integrations
 - Power Platform
 - Security
 - Intelligent performance
 - Monitoring
 - Automation
 - Help

Essentials

Resource group (move)	: EventEaseApp-rg	Server name	: eventeaseapp-server-db.database.windows.net
Status	: Online	Elastic pool	: No elastic pool
Location	: Austria East	Connection strings	: Show database connection strings
Subscription (move)	: Azure for Students	Pricing tier	: Basic
Subscription ID	: 06e4c813-d342-49ef-9e3a-3315fd8afd6d	Earliest restore point	: 2026-03-16 21:44 UTC
Tags (edit)	: Add tags		

Getting started

Monitoring Properties Features Notifications (0) Integrations Tutorials

Start working with your database

Connect to your database and start working with data with a few simple steps. [Learn more](#)



Configure access

Configure network access to your SQL server. [Learn more](#)

Configure



Connect to application

Use connection strings to connect to your SQL database from your applications and favorite tools.

See connection strings



Start developing

Work in your database by using tools to add, modify and query data. [Compare tools](#)

Open in Visual Studio Code

Open in SQL Server Management Studio



Mirror database in Fabric

Replicate existing databases in Fabric, and help your team achieve streamlined ETL and operational analytics goals. [Learn more](#)

CLDV7111_database (eventeaseapp-server-db/CLDV7111_database) | Query editor (preview) ☆ ...

Search

New query Templates Open in Feedback Sign out Classic experience

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Views...Stored ProceduresFunctions

Queries

SQL query 1

SQL query 1

Run Save as view

Your queries will be lost when your session ends. Use Save as view to keep results as a database view, or copy your query text to save it externally.

1 SELECT * FROM Venues

MessagesResultsDownloadCodeDownload

	123	VenueId	ABC	VenueName	ABC	Location	123	Capacity	ABC	ImageUrl
1	1			moes mabhida		durban		1222		https://example
2	2			Playhouse		Durban		2000		https://playhouse.com

Succeeded (0 sec 295 ms)Columns: 5 Rows: 2

Explorer

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dbo

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_EFMigrationsBookingsEventsVenues

Views...Stored ProceduresFunctions

Queries

SQL query 1

SQL query 1

Run Save as view

Your queries will be lost when your session ends. Use Save as view to keep results as a database view, or copy your query text to save it externally.

1 SELECT * FROM Bookings

MessagesResultsDownloadCodeDownload

	123	BookingId	123	EventId	123	VenueId	ABC	BookingDate
1	1		1		2			2026-03-27T15:13:00.0000000

Succeeded (0 sec 294 ms)Columns: 4 Rows: 1

New query Templates Open in Feedback Sign out Classic experience

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Views...Stored ProceduresFunctions

Queries

SQL query 1

SQL query 1

Run Save as view

Your queries will be lost when your session ends. Use Save as view to keep results as a database view, or copy your query text to save it externally.

1 SELECT * FROM Events

MessagesResultsDownloadCodeDownload

	123	EventId	ABC	EventName	ABC	EventDate	ABC	Description	123	VenueId
1	1			SummerBreeze		2026-03-26T14:43:00.0000000		svcsgrfdkj		1
2	2			Zwide		2026-03-25T15:15:00.0000000		Zwide's event		1

Succeeded (0 sec 294 ms)Columns: 4 Rows: 1

2.4 Azure Resource Group

portal.azure.com/#@advtechonline.onmicrosoft.com/resource/subscriptions/06e4c813-d342-49ef-9e3a-3315fd8af6d/resourceGroups/EventEaseApp-rg/overview

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EventEaseApp-rg Resource group

Export resource groups using Bicep or Terraform How to manage changes with deployment tools? Help me generate Terraform for this resource group configuration.

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete Export template Open in mobile Add to service group Group by none

Overview

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Essentials

Subscription (move) : Azure for Students Deployments : 2 Succeeded

Subscription ID : 06e4c813-d342-49ef-9e3a-3315fd8af6d Location : West US 2

Tags (edit) : Add tags

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name	Type	Location
CLDV7111_database (eventeaseapp-server-db/CLDV7111_database)	SQL database	Austria East
EventEaseApp	App Service	Austria East
eventeaseapp-server-db	SQL server	Austria East
EventEaseAppPlan	App Service plan	Austria East

Showing 1 - 4 of 4. Display count: auto Give feed

GitHub:

github.com/MnqobiNxumalo/EventEaseApp/tree/master/EventEaseApp/Controllers

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MnqobiNxumalo / EventEaseApp

Type to search

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Files

master

Go to file

EventEaseApp

Controllers

BookingsController.cs

EventsController.cs

HomeController.cs

VenuesController.cs

Data

ApplicationDbContext.cs

DesignTimeDbContextFactory.cs

Models

Booking.cs

ErrorViewModel.cs

Event.cs

EventEaseApp / EventEaseApp / Controllers

Add file

Student and Student Add project files. 544a77e · 4 minutes ago History

Name	Last commit message	Last commit date
..		
BookingsController.cs	Add project files.	4 minutes ago
EventsController.cs	Add project files.	4 minutes ago
HomeController.cs	Add project files.	4 minutes ago
VenuesController.cs	Add project files.	4 minutes ago

<https://github.com/MnqobiNxumalo/EventEaseApp>

3. Cloud Computing Basics

3.1

- Security:
 - On-Premises: The company has complete digital and physical control. You are in charge of everything, including server patching and creating access. This can be safe, but it comes with a significant cost and responsibility. For instance, because of stringent regulations, a bank may decide to keep all sensitive financial data on-site.
 - Cloud: We all share responsibilities for security. The actual infrastructure (hardware, data centers) is secured by the cloud provider, Microsoft. The company is in charge of protecting its data, application configuration, and user access (passwords, MFA). Example: EventEaseApp can rely on Azure's built-in threat monitoring and firewalls, but they still need to make sure their booking experts use secure passwords.
- Deployment Speed:
 - On-Premises: Slow. Hardware acquisition, setup, configuration, and networking are all part of a protracted cycle that can take weeks or months. For example, placing an order for a new server, awaiting arrival, racking it, setting up the operating system, and then launching the program.
 - Cloud: Quick. Resources are API-driven and virtual. Using a script or a gateway, you may launch a new database or web application in a matter of minutes. Example: EventEaseApp doesn't need to wait for new hardware in order to scale up their App Service plan during peak event season with just a few clicks.
- Resource Management:
 - On-Premises: Needs capacity planning and is rigid. Hardware must be purchased in sufficient quantities for peak loads, which are frequently idle. The procedure of upgrading is labor-intensive and laborious.

- Cloud: flexible. Resources may be immediately scaled up or down in response to demand, and you only pay for what you use. This reduces capacity waste and streamlines administration. For Example, Azure can automatically add more resources to the App Service if EventEaseApp's traffic increases; when traffic decreases, it scales back down, saving money.

3.2.

- Definitions:

- ❖ Infrastructure as a Service, or IaaS, offers on-demand access to servers, virtual machines, storage, and networks. You oversee the OS, middleware, and applications while renting the infrastructure. Azure Virtual Machines is one example
- ❖ Platform as a Service, or PaaS, offers a platform for creating, executing, and managing applications without the hassles of developing and maintaining the underlying infrastructure (servers, operating system, etc.). You stare at the code. Example: Azure SQL Database and Azure App Service.

Software as a Service, or SaaS, offers operational software via the internet. Everything is managed by the vendor. For instance, Salesforce and Microsoft Office 365.

- Why PaaS is Best for EventEaseApp:

- Decreased Management Overhead: EventEaseApp doesn't have to bother about deploying SQL Server updates, maintaining IIS, or patching the server OS while using PaaS (Azure App Service & Azure SQL). The development team can concentrate just on creating the booking application as Microsoft takes care of that.
- Speed and Efficiency: Code can be deployed straight from Visual Studio. Scaling the app is easy and integrated. For a "up-and-coming" business, this speed is essential.

- Comparison:

- In order to implement IaaS, they would need to install and configure SQL Server and IIS, manage the operating system, provision a virtual machine,

and take care of all upgrades and security patches. This is more difficult and time-consuming than what a business application needs.

- SaaS would be a ready-made reservation system. Although simple, it lacks the unique features that EventEase requires (such as loading events before a venue is ready or prohibiting duplicate bookings using their particular logic). PaaS is the ideal middle ground because they require a custom solution, which necessitates a platform to develop upon.